

Homework # 4-Solutions Set

1. (80 points) Let the function **fun** be defined as follows:

```
int fun(int i, int *j){
    i *= 2;
    *j *= 2;
    return i * (*j) + 1;
}
```

Suppose **fun** is used in a **main** program as follows:

```
void main(){
    int x, y, z, v, result1, result2, result3, result4;
    x = 5;
    y = 7;
    z = 5;
    v = 7;
    result1 = x + y + fun(x,&y);
    printf("Result1 = %d\n", result1);
    printf("x = %d\n", x);
    printf("y = %d\n", y);

    result2 = fun(z,&v) + z + v;
    printf("Result2 = %d\n", result2);
    printf("z = %d\n", z);
    printf("v = %d\n", v);

    result3 = x + y * fun(x,&y);
    printf("Result3 = %d\n", result3);
    printf("x = %d\n", x);
    printf("y = %d\n", y);

    result4 = fun(z,&v) * z + v;
    printf("Result4 = %d\n", result4);
    printf("z = %d\n", z);
    printf("v = %d\n", v);
}
```

Which parameter passing methods are used for each parameter? Explain.

What will be printed on the screen?

a) If the operands in the expressions are evaluated from left to right?

Answer: If the operands are evaluated from left to right, the result values are computed as below:

Result1 = $5 + 7 + (10 * 14 + 1) = 153$

Result2 = $(10 * 14 + 1) + 5 + 14 = 160$

Result3 = $5 + 14 * (10 * 28 + 1) = 3939$

Result4 = $(10 * 28 + 1) * 5 + 28 = 1433$

So, the below values are printed on the screen:

```
Result1 = 153
x = 5
y = 14
Result2 = 160
z = 5
v = 14
Result3 = 3939
x = 5
y = 28
Result4 = 1433
z = 5
v = 28
```

b) If the operands in the expressions are evaluated from right to left?

Answer: If the operands are evaluated from right to left, the result values are computed as below:

```
Result1 = 5 + 14 + (10 * 14 + 1) = 160
Result2 = (10 * 14 + 1) + 5 + 7 = 153
Result3 = 5 + 28 * (10 * 28 + 1) = 7873
Result4 = (10 * 28 + 1) * 5 + 14 = 1419
```

So, the below values are printed on the screen:

```
Result1 = 160
x = 5
y = 14
Result2 = 153
z = 5
v = 14
Result3 = 7873
x = 5
y = 28
Result4 = 1419
z = 5
v = 28
```

c) If this program is compiled and run in gcc (or DevC)?

Answer: The output is given in the below screen shot.

```
D:\Files\CU\programming languages\2014-2015\Proje1.exe
Result1 = 153
x = 5
y = 14
Result2 = 160
z = 5
v = 14
Result3 = 7873
x = 5
y = 28
Result4 = 1433
z = 5
v = 28

-----
Process exited after 0.02177 seconds with return value 0
Devam etmek için bir tuşa basın . . .
```

d) If this program is compiled and run in C and C++ compiler of Microsoft Visual Studio?

Answer: The output is given in the below screen shot.

```
C:\Windows\system32\cmd.exe
Result1 = 153
x = 5
y = 14
Result2 = 160
z = 5
v = 14
Result3 = 7873
x = 5
y = 28
Result4 = 1433
z = 5
v = 28
Devam etmek için bir tuşa basın . . .
```

e) Explain the results obtained in parts c and d.

Answer: Both Dev-C and C++ compiler in Microsoft Visual Studio provided the same output. However, these outputs are different from the computed result values in parts a and b. Both compilers evaluate operands from left to right, when the operand has higher priority and one of the operands of this operator (*) in this question) has function call, first function call is computed, after that the other operands are evaluated from left to right.

2. (20 points) Rewrite the following code using no gotos or breaks.

```

int i = -4;
for (j = 0; j < 4; j++){
    switch (i + 3) {
        case 4:
            case 3: i--; break;
            case 1: i += 2; break;
            default: i = 0;
        }
        if (i > 0) break;
        i = 4 - j;
    }
}

```

Answer:

```

int i = -4;
for (j = 0; (j < 4) && (i <= 0); j++){
    if (((i + 3) == 4) || ((i + 3) == 3))
        i--;
    else if ((i + 3) == 1)
        i += 2;
    else
        i = 0;

    i = 4 - j;
}

```